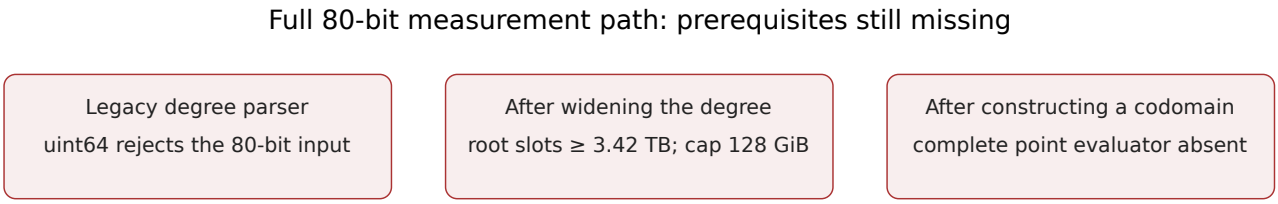
Requested gap80 measurements: execution prerequisites

The native metadata audit confirms why the existing constructor cannot yet measure the requested complete 80-bit crossing. No performance experiment was launched, and no missing time is presented as zero or extrapolated.



Exact input and observed eligibility checks

The frozen degree is 729845429276318409687041 (80 bits). The earlier `sqrt_velu_batched_v4.cpp` parses it into uint64; the audit reproduces that extraction and observes rejection. Its scalar multiplication also uses that width. A prior 96-bit field receipt is not an 80-bit degree receipt.

With a widened degree and the same window rule, the exact baby count is 427,154,957,034. Each fmpz slot occupies 8 bytes in this build; their array alone needs 3,417,239,656,272 bytes. This excludes integer limbs, derivative polynomials and product/remainder trees. It exceeds the 137,438,953,472-byte lab cap. This is a method-specific accounting bound, not an observed OOM or a lower bound for all isogeny algorithms.

The earlier constructor's source and CONSTRUCTOR.md explicitly limit it to codomain construction. Arbitrary-point evaluation and a transported base are absent. Consequently paired endpoint logarithm timings cannot be formed from the current artifacts.

Provenance, controls and next executable

`../native/measurement_preflight.cpp` implements exact GMP metadata checks; `../measurement-readiness.json` preserves their output. The degree-10,009 control accepts the legacy parser and needs 50 root slots (400 bytes), without claiming the metadata check constructs an isogeny. The build script compiles this tool without launching a crypto performance experiment.

`../MEASUREMENT_PROTOCOL.md` preserves the requested complete construction, 32 non-kernel contexts, scalar multiplication and matched ECDLP trials, including setup/online/verification boundaries, controls and the earlier index-calculus request. Iso Lab EKS/native execution and checkpoints remain required. No fallback fixture was dispatched in place of the target.

Next: implement an 80-bit-degree-compatible constructor with feasible total memory/work and a verified full point evaluator; then freeze its source/image/binary, scalar list and budget and run the paired protocol. Source-only solves and partial tiles remain separate diagnostics. See `problems/conductor-gap-upper-floor-and-cost.md` for the continuing obligation.